

TNBBeta-VAE Research Summary: Week 2

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1 Overview

Following last week's empirical discovery that the TNBBeta-VAE posteriors closely resembled the S-VAE's posteriors¹, the primary goals for this week were:

1. Understand why the posteriors converge to each other
2. Determine whether there are situations in which they do not

I additionally included the power-spherical VAE [Cao and Aziz, 2020] in this week's comparisons. We discovered two clear wins for the TNBBeta-VAE last week: this model buys us significantly faster sampling over the S-VAE while admitting stable gradient calculation in high dimensions. The power-spherical VAE has the exact same benefits, while additionally having a closed-form KL divergence to the uniform prior². If we were to find a scenario in which the TNBBeta-VAE additionally outperforms the power-spherical VAE on a particular task, then we could paint a more complete picture on how this model improves upon existing hyperspherical approaches to VAEs.

As we will see in section 2, I was able to answer both questions (1) and (2). In the limit $p \rightarrow 1$, under a specific condition on ε , the SphericalTNBBeta density and vMF density converge to dirac masses about the mean direction μ . Along this path, they both admit a cap-like shape about μ , though, crucially, their tails are fundamentally different in mass. To answer question (2), I trained a series of graph-VAEs on com-DBLP [Yang and Leskovec, 2012] in hopes that complex community relationships could tease out the additional expressivity of the SphericalTNBBeta. I found that the TNBBeta-VAE significantly outperformed both the S-VAE and the power-spherical VAE, and this performance was especially pronounced for nodes with edges connected to five or more communities. For this particular task, the SphericalTNBBeta posterior actually took on a bimodal shape that the vMF and power-spherical distributions can never attain.

¹recall: cap-shaped structure on the sphere.

²we use MC estimates for the TNBBeta-VAE.

2 Results

First, let's discuss convergence.

Theorem 2.1 (Modality of SphericalTNBBeta). *Fix $\mu \in \mathbb{S}^{d-1}$. For any $p \in (0, 1)$ and $q \in [0, 1]$:*

- *if $\varepsilon > (d - 1)/2$, SphericalTNBBeta(μ, p, q, ε) is unimodal;*
- *if $\varepsilon < (d - 1)/2$, SphericalTNBBeta(μ, p, q, ε) admits two modes at μ and $-\mu$, and may admit a third interior mode depending on the concentration parameter q .*

Theorem 2.2 (Convergence of SphericalTNBBeta). *Fix $\mu \in \mathbb{S}^{d-1}$, $q \in [0, 1]$, and $\varepsilon > (d - 1)/2$. As $p \rightarrow 1$, SphericalTNBBeta(μ, p, q, ε) converges to δ_μ and mass is concentrated within $O(1 - p)$ of μ*

The vMF distribution always has a cap-shape, and trivially converges to a dirac mass about μ as $\kappa \rightarrow \infty$.

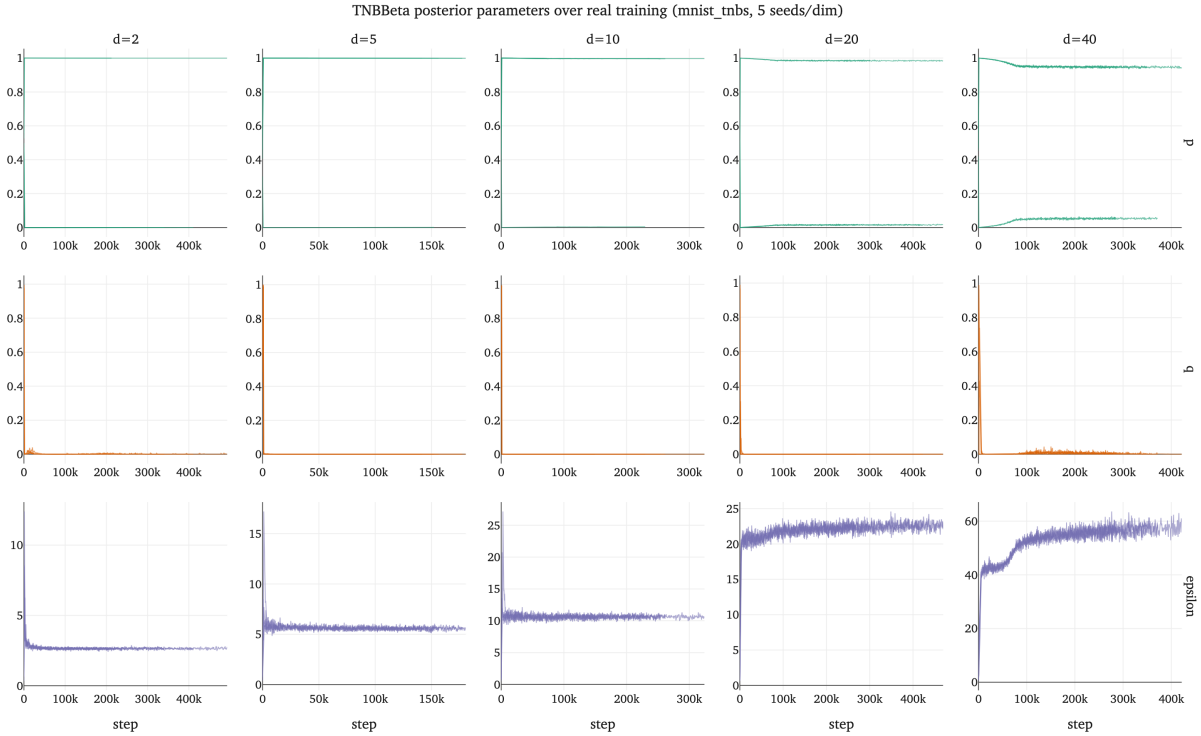


Figure 1: Posterior trajectories for the SphericalTNBBeta-VAE

Figure 1 illustrates that the limit in Theorem 2.2 is actually met in practice, validating the coarse "cap-like" shape argument made in week 1.

I additionally extended the MNIST evaluation table from week 1 to include the power-spherical VAE:

d	S-VAE (vMF, $\mathbb{S}^d$)				TNBBeta ($\mathbb{S}^d$)				Power Spherical ($\mathbb{S}^d$)			
	LL	L[q]	RE	KL	LL	L[q]	RE	KL	LL	L[q]	RE	KL
2	-130.86 $\pm$ 0.43	-134.42 $\pm$ 0.42	-126.89 $\pm$ 0.47	7.53 $\pm$ 0.08	-130.11 $\pm$ 0.57	-134.21 $\pm$ 0.46	-126.43 $\pm$ 0.54	7.78 $\pm$ 0.11	-131.02 $\pm$ 0.39	-134.56 $\pm$ 0.28	-127.03 $\pm$ 0.40	7.53 $\pm$ 0.13
5	-105.27 $\pm$ 0.14	-109.32 $\pm$ 0.08	-95.56 $\pm$ 0.12	13.76 $\pm$ 0.06	-104.89 $\pm$ 0.22	-109.15 $\pm$ 0.22	-95.36 $\pm$ 0.26	13.79 $\pm$ 0.13	-105.37 $\pm$ 0.17	-109.38 $\pm$ 0.22	-95.67 $\pm$ 0.22	13.71 $\pm$ 0.08
10	-89.38 $\pm$ 0.02	-93.60 $\pm$ 0.06	-71.84 $\pm$ 0.08	21.77 $\pm$ 0.05	-89.29 $\pm$ 0.10	-93.65 $\pm$ 0.14	-71.96 $\pm$ 0.25	21.69 $\pm$ 0.14	-89.42 $\pm$ 0.10	-93.69 $\pm$ 0.09	-71.97 $\pm$ 0.10	21.71 $\pm$ 0.03
20	-86.84 $\pm$ 0.04	-92.10 $\pm$ 0.05	-62.63 $\pm$ 0.21	29.47 $\pm$ 0.18	-86.90 $\pm$ 0.23	-92.12 $\pm$ 0.15	-62.33 $\pm$ 0.41	29.78 $\pm$ 0.53	-86.95 $\pm$ 0.08	-92.17 $\pm$ 0.07	-62.72 $\pm$ 0.18	29.45 $\pm$ 0.15
40	-144.28 $\pm$ 31.06	-170.06 $\pm$ 41.29	-162.32 $\pm$ 56.08	7.73 $\pm$ 14.79	-89.09 $\pm$ 0.15	-96.38 $\pm$ 0.14	-62.53 $\pm$ 0.42	33.86 $\pm$ 0.36	-89.04 $\pm$ 0.11	-96.37 $\pm$ 0.13	-62.17 $\pm$ 0.23	34.20 $\pm$ 0.23

Table 1: MNIST Table 1, same as last week, replacing the Gaussian results with the power-spherical VAE.

Importantly, there is no significant difference between any of the hyperspherical methods, except at $d = 40$, in which the S-VAE becomes unstable as we saw last week. Though the power-spherical VAE fixes this problem, it achieves similar results to the TNBBeta-VAE.

Moving on to the graph representation task, I ran experiments with both the TNBBeta-VAE and S-VAE on PubMed last week, but saw nearly identical results with both posteriors. My hypothesis was that that dataset had too few connected communities, and as such a unimodal posterior was sufficient to describe the latent space. The Stanford DBLP dataset is significantly more diverse. It’s a co-authorship network built on research papers in computer science. Each node is an author, and they are connected if they have published at least one paper together. Ground-truth communities are defined by publication venue. Per Stanford’s dataset statistics, the graph has a clustering coefficient of ≈ 0.63 , indicating that there are many connected communities, indicating that a VAE with a multimodal latent could aid in graph representation problems.

To test this hypothesis out, I trained a graph-VAE on com-DBLP, and evaluated link prediction (i.e. predicting whether two held-out nodes are actually connected). Note that there are no node features in this particular dataset. To assess performance, I evaluated AUC and AP both on the entire test set, as well as buckets of the test set grouped by the number of communities the contained nodes belong to. We see that the TNBBeta-VAE outperforms both the S-VAE and the power-spherical VAE:

Metric	TNBBeta	Power Spherical	vMF	TNB vs. PS	TNB vs. vMF
Primary AUC	0.9205 $\pm$ 0.0120	0.8901 $\pm$ 0.0122	0.8794 $\pm$ 0.0180	$p = 0.0074$	$p = 0.0068$
Primary AP	0.9237 $\pm$ 0.0095	0.8891 $\pm$ 0.0113	0.8810 $\pm$ 0.0165	$p = 0.0017$	$p = 0.0036$
Secondary AUC	0.8610 $\pm$ 0.0200	0.8212 $\pm$ 0.0183	0.8059 $\pm$ 0.0253	$p = 0.0192$	$p = 0.0098$
Secondary AP	0.7816 $\pm$ 0.0222	0.7033 $\pm$ 0.0252	0.6884 $\pm$ 0.0342	$p = 0.0017$	$p = 0.0027$

Table 2: Link-prediction AUC/AP (mean $\pm$ std, 5 seeds), aggregated over the whole test set and broken down by a bridge node’s primary vs. secondary community. p -values are Welch’s t -test.

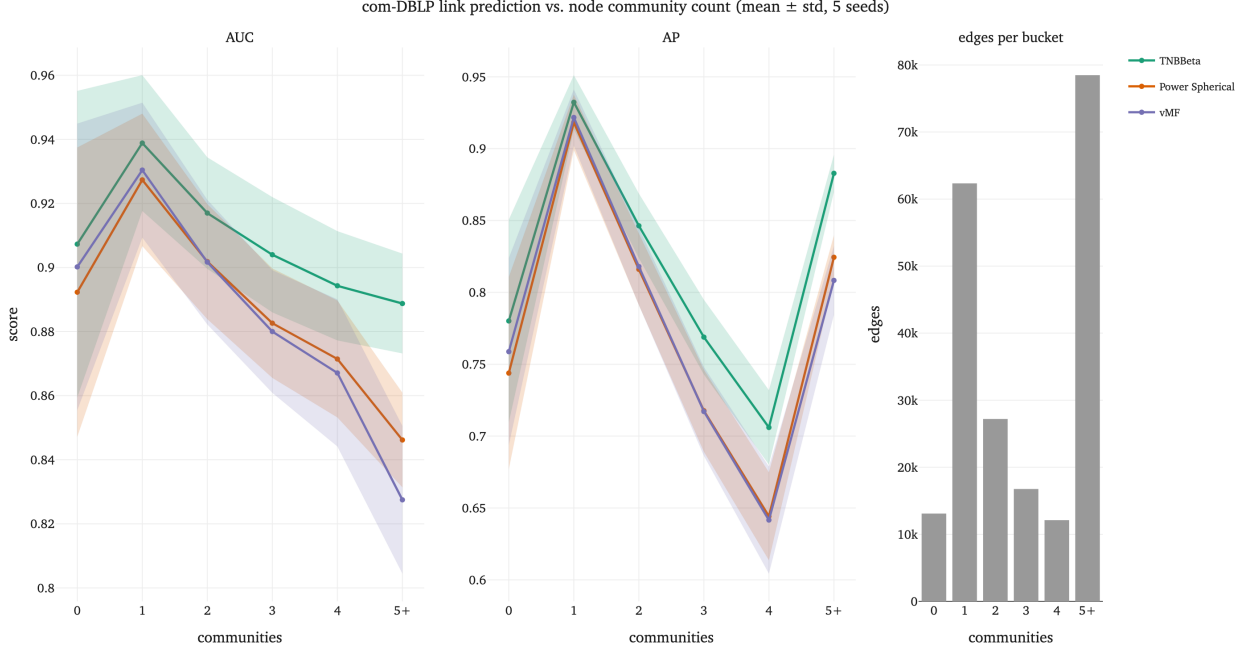


Figure 2: Link-prediction AUC and AP by community bucket.

It is important to note that in all cases, the SphericalTNBBeta posterior always had $\varepsilon < (d-1)/2$, meaning that this particular task leveraged multimodality of the posterior. [Theorem 2.3](#) shows that this cannot possibly hold for the vMF or power-spherical distributions:

Theorem 2.3 (Unimodality of vMF and Power Spherical). *For every $\mu \in \mathbb{S}^{d-1}$ and every $\kappa > 0$, the $\text{vMF}(\mu, \kappa)$ and $\text{PowerSpherical}(\mu, \kappa)$ distributions are unimodal, with unique mode at μ*

3 Open Questions

I don’t exactly have any “open questions” this week, per se, so I’ll take the time to use this section as a space for reflections. It seems as though, when the latent space is “simple” enough (e.g. MNIST), the TNBBeta-VAE is comfortable ditching its multimodal capacity. When the space gets more complicated (i.e. representing graphs with several connected communities), the model adapts and uses its capacity for expressivity. The benefit here is that one need not consider whether a unimodal or multimodal posterior is preferable when making a model selection decision; in the unimodal case, the TNBBeta-VAE performs just as well as the power-spherical VAE anyway. I see this model as a strict improvement, at the cost of an MC-estimated KL-divergence.

I’d be curious in exploring other datasets in the wild that need to leverage multimodality. The fact that two of the SphericalTNBBeta modes are located at opposite poles of the unit hypersphere gives some signal here. A dataset such as positive/negative sentiment reviews could be useful in that respect.

Acknowledgement of Generative AI Use

Generative AI (specifically, Claude Sonnet 5) was used in the generation of all code to assist with running the above experiments, as well as generating the above visualizations. It was additionally used to assist with proof-reading the proofs of theorems 2.1 and 2.3, as well as helping with a sketch of theorem 2.2. I assert that I have reviewed the above results and claim responsibility for the above statements. All commentary in the writeup aside from the captions is my own. So is all analysis above.

A Proofs

Proof of Theorem 2.1. Recall the definition of the SphericalTNBBeta density from week 1's summary. We may proceed closely as in the proof of proposition 3.1 of [Lederman and Schein \[2026\]](#), and write

$$f(y) \stackrel{\text{def}}{=} \text{SphericalTNBBeta}(y; \mu, p, q, \varepsilon) = \text{Beta}(y; \varepsilon, \varepsilon) [y(1-y)]^{-(\frac{d-3}{2})} h(y)$$

The function h above is continuous, strictly positive, and bounded on $[0, 1]$, as argued in proposition 3.1. We now consider a slightly modified kernel:

$$K(y) = \frac{1}{B(\varepsilon, \varepsilon)} [y(1-y)]^{\varepsilon - \frac{d-1}{2}}$$

The remainder of the proof is almost identical to that of proposition 3.1, with the exception that we now consider the cases $\varepsilon > (d-1)/2$, $\varepsilon = (d-1)/2$, and $\varepsilon < (d-1)/2$. $\square$

Proof of Theorem 2.2. Consider the family of random variables $\{Z_p\}_{p \in (0,1)}$ where

$$Z_p \sim \text{SphericalTNBBeta}(\mu, p, q, \varepsilon)$$

We know that for each p , the latitude Y_p is distributed $\text{TNBBeta}(p, q, \varepsilon)$. Let $T_p \stackrel{\text{def}}{=} (1 - Y_p)/(1 - p) = (1 - Y_p)/\delta$. It suffices to show that the family $\{T_p\}$ is tight. By change of variables, the density for T_p is given by

$$\begin{aligned} f_\delta(t) &= \delta \frac{(1-\delta t)^{\varepsilon-1} (\delta t)^{\varepsilon-1}}{B(\varepsilon, \varepsilon)} \left[\frac{1-q}{(1-\delta t)\delta t} \gamma \right]^\varepsilon [1-4q\gamma]^{-(\varepsilon+\frac{1}{2})} \\ &= \frac{1}{t(1-\delta t)B(\varepsilon, \varepsilon)} [(1-q)\gamma]^\varepsilon [1-4q\gamma]^{-(\varepsilon+\frac{1}{2})} \\ &= \frac{(1-q)^\varepsilon}{B(\varepsilon, \varepsilon)} \cdot [1-4q\gamma]^{-(\varepsilon+\frac{1}{2})} \cdot \frac{\gamma^\varepsilon}{t(1-\delta t)} \end{aligned}$$

Because of the sech term, it follows that $\gamma \leq 1/4$ regardless of the value of δ . We then see that

$$f_\delta(t) \leq \frac{(1-q)^\varepsilon}{B(\varepsilon, \varepsilon)} (1-q)^{-(\varepsilon+\frac{1}{2})} \frac{\gamma^\varepsilon}{t(1-\delta t)} = \frac{1}{\sqrt{1-q}B(\varepsilon, \varepsilon)} \frac{\gamma^\varepsilon}{t(1-\delta t)}$$

Now, we rewrite γ :

$$\gamma = \frac{\delta t(1-\delta t)(1-\delta)\delta}{[(1-\delta)\delta t + \delta(1-\delta t)]^2} = \frac{t(1-\delta t)(1-\delta)}{[t(1-\delta) + 1 - \delta t]^2}$$

and we can see that for $\delta < 1/2$,

$$\frac{1}{[t(1-\delta) + 1 - \delta t]^2} \leq \frac{1}{[1 + t(1-2\delta)]^2} \leq \frac{1}{[1 + ct]^2} \leq \frac{4}{t^2}$$

Using the fact that $(1-\delta) < 1$ we bound γ above as

$$\gamma \leq \frac{4(1-\delta t)}{t}$$

Hence, substituting this inequality for γ back into f_δ yields

$$f_\delta(t) \leq \underbrace{\frac{4^\varepsilon}{\sqrt{1-qB(\varepsilon, \varepsilon)}}}_K t^{-\varepsilon-1} (1-\delta t)^{\varepsilon-1} \leq K t^{-\varepsilon-1}$$

It then follows that for any $\zeta > 0$, choose $M = (\zeta K/\varepsilon)^{(1/\varepsilon)}$ such that

$$\mathbb{P}\{T_p > M\} \leq K \int_M^{1/\delta} t^{-\varepsilon-1} dt < K \int_M^\infty t^{-\varepsilon-1} dt = \frac{K}{\varepsilon M^\varepsilon} = \zeta$$

and so the family $\{T_p\}$ is tight since p was chosen arbitrarily above.

It immediately follows from tightness of T_p that Z_p is concentrated within $O(1-p)$ about μ . Now, we see that for any ζ , as $p \rightarrow 1$,

$$\mathbb{P}\{1 - Y_p > \zeta\} = \mathbb{P}\left\{T_p > \frac{\zeta}{1-p}\right\} \rightarrow 0$$

It follows that $Y_p \rightarrow 1$ in probability, and thus $Z_p \rightarrow \mu$ in probability. $\square$

Proof of Theorem 2.3. This largely follows from definitions 1 and 2 of Cao and Aziz [2020]. Fix $\mu \in \mathbb{S}^{d-1}$, $\kappa > 0$. The unnormalized vMF and power-spherical densities are given by

$$p_{\text{vMF}}(\mathbf{x}; \mu, \kappa) \propto \exp\{\kappa \mu^\top \mathbf{x}\}, \quad p_{\text{PS}}(\mathbf{x}; \mu, \kappa) \propto (1 + \mu^\top \mathbf{x})^\kappa$$

Since $\kappa > 0$, observe that $\exp$ and $y \mapsto y^\kappa$ are monotone increasing in $\mu^\top \mathbf{x}$, which is uniquely maximized (equal to 1) by $\mathbf{x} = \mu$. Hence, both densities admit a unique mode at $\mathbf{x} = \mu$. $\square$

References

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